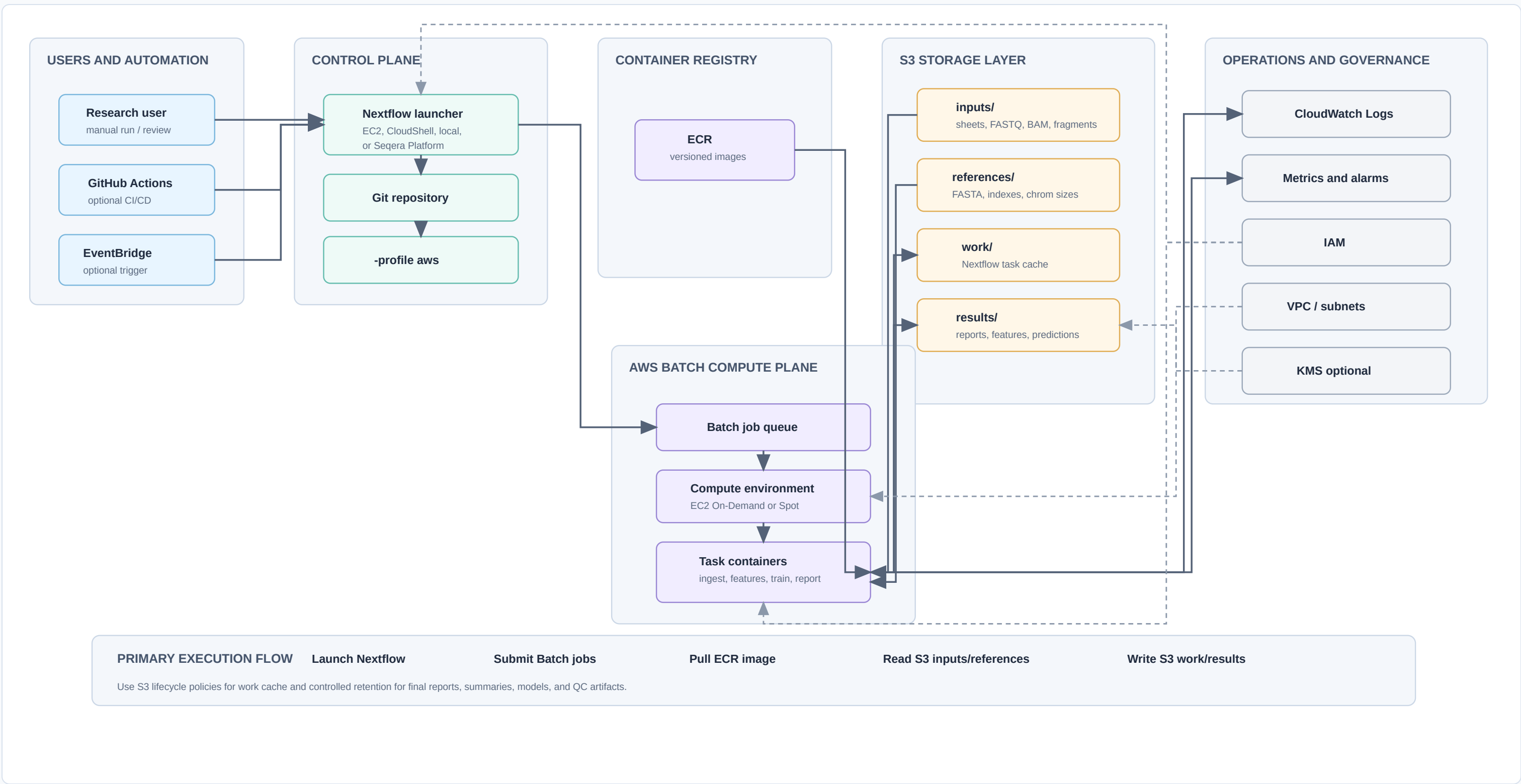


Fragmentomics NF AWS Cloud Architecture

Reference deployment pattern for running the pipeline with Nextflow, AWS Batch, S3, ECR, CloudWatch, IAM, and VPC controls.

Potential AWS reference architecture | research workflow



Compute pattern

Nextflow remains the orchestration layer and submits each process as an AWS Batch job using the `awsbatch` executor.

Storage pattern

S3 holds input data, references, the Nextflow work directory, and final results. Keep work cache lifecycle separate from curated outputs.

Governance pattern

IAM, VPC placement, CloudWatch logging, and optional KMS encryption provide access control, isolation, and operational traceability.